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An evaluation of the prevalence and causes of blood donor deferral in Lagos state, Nigeria

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Abstract

Background Blood transfusion safety relies on thorough donor eligibility assessments that lead to the deferral of individuals who do not meet established criteria. This study aimed to evaluate the rates and reasons for blood donor deferral in Lagos.

Methods A five-year retrospective analysis of donor records from January 2017 to December 2021 was conducted at two tertiary and one secondary blood transfusion service centres in Lagos. Statistical analyses assessed the causes of deferral across sex and age categories.

Results Among 47,473 registered donors, 11,484 (24.2%) were deferred, with females experiencing a significantly higher deferral rate (37.4%, $p < 0.0001$) than males (19.2%). Temporary deferrals were predominant (82.3%), with low haemoglobin (20.9% of females and 4.1% of males) as the leading cause. Hepatitis B Virus (HBV) infection was the most frequent reason for permanent deferral, affecting 3.5% of males and 0.3% of females. Age distribution analysis revealed the highest deferral rate in the 16–25 age group (26.1%).

Conclusion Nearly a quarter of potential donors were deferred. This high deferral rate contributes to blood shortages. Public health initiatives that target anaemia, nutritional deficiencies, and HBV infection, along with enhanced donor education and follow-up programmes, are crucial for increasing the number of eligible donors.

Keywords Blood donor, Donor deferral, Donor eligibility, Donor selection criteria, Temporary and permanent deferral

1 Introduction

Ensuring a safe and adequate supply of blood and its products is one of the major challenges facing blood transfusion services (BTSs) worldwide. A significant imbalance exists in access to safe blood between developing and developed countries [1]. Nigeria and other sub-Saharan African countries face severe blood supply deficits, with consequent negative influence on the quality of healthcare delivery [2–4].

Although an adequate supply of blood is paramount in blood transfusion practice, it is also crucial to ensure that the collection process does not pose a risk to donor health [5] and that suitable donors are not unnecessarily deferred [6]. This is achieved by

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implementing stringent donor eligibility criteria based on medical evidence and current guidelines [7, 8]. Donors who do not meet set eligibility criteria are deferred, which may involve a temporary postponement or permanent exclusion from blood donation [6].

Approximately 13 million potential donors globally are deferred annually due to suspected or confirmed haematological disease, infectious diseases, or other medical conditions that may jeopardise donor safety and recipient health [9, 10].

The incidence of blood donor deferrals varies worldwide, with rates ranging from 7 to 12.7% reported in some parts of Africa and 4.6–30% reported in Asia [11, 12]. There is a need to assess the magnitude of donor deferrals continually. More eligible donors will increase blood supplies and vice versa. Understanding the donor deferral patterns will guide strategies for effective donor selection, promote donor safety, optimise blood stock, and reduce wastage. Hence, this study aimed to analyse the prevalence and causes of blood donor deferrals at selected blood transfusion service sites in Lagos, Nigeria.

2 Methods

Study design and settings: This retrospective study was conducted at three selected donor clinics: Lagos State Blood Transfusion Service (LSBTS) Gbagada Lagos, National Orthopaedic Hospital Igbobi, Lagos and General Hospital Mushin, Lagos. It involved a review of the data of all donors who presented at these sites between January 1st, 2017, and December 31st, 2021. The blood donor eligibility criteria adopted by the institutions were per the Association for the Advancement of Blood and Biotherapies (AABB) and World Health Organisation (WHO) blood donor selection and counselling guidelines [6].

Five-year data from the institutions' donor registers were reviewed and analysed to evaluate the causes and prevalence of donor deferral. The relevant information retrieved for review and analysis included the age of blood donors, sex, vital signs, haematocrit/haemoglobin value, body weight, temperature, frequency of donation, pre-donation transfusion transmissible infection (TTI) screening results, and other donor selection criteria adopted by the institutions.

The donor deferral criteria used by the institutions for permanent deferral included overage, chronic health conditions, and seropositivity for any TTIs - human immunodeficiency virus (HIV), hepatitis B virus (HBV), hepatitis C virus (HCV), and syphilis in centres where pre-transfusion screening was carried out. Other factors such as haemoglobin level of less than 12.5 g/dl and/or a haematocrit value of less than 38%, recent donation of shorter than three months for males and four months for females, body weight less than 50 kg, alcohol intake within 24 h, presence of skin piercing and tattoos, abnormal vital signs, pulse rate (value outside 60–100 per minute with or without irregularities), temperature above 37.6 °C, blood pressure (systolic and diastolic blood pressure outside;120–129 and 80–89, respectively), asthma were categorised as criteria for temporary deferral.

The serological tests for TTIs (HIV, HBV, HCV, and syphilis) were carried out via a combination of pre-donation rapid diagnostic testing (RDT) and enzyme-linked immunosorbent assays (ELISAs) post-donation. The pre-donation testing, however, was not performed at all study sites.

According to donor site protocol, all prospective donors were registered and counselled, and eligibility for blood donation was assessed using a standardised donor

questionnaire. Measurements of weight, haemoglobin levels, and blood pressure were then taken. Donors who do not fulfil the eligibility criteria are either temporarily or permanently deferred based on their clinical findings. At sites where pre-donation TTI screening was conducted, samples were collected and the tests performed using RDT kits. Those with reactive TTI results were notified and counselled confidentially by a qualified medical officer or counsellor. Counselling included explanation of test results, education on the nature of the infection, emotional support, and referrals for confirmatory testing and medical follow-up at a designated facility. Donors who met all eligibility criteria proceeded to donate blood after giving informed consent.

The determination of haemoglobin level was performed using either the copper sulphate solution or the haemocyte method, while packed cell volume/haematocrit was determined via the microhaematocrit method (centrifugation method).

2.1 Statistical analysis

Data from the donors recruited at the centres were collected and entered into a Microsoft Excel spreadsheet for analysis. Descriptive statistics were presented as frequencies and proportions. The causes of deferral were grouped and analysed. The associations of categorical variables such as sex and causes of donor deferral were tested via chi-square analysis. A statistically significant difference was set at an alpha value of 5% ($p < 0.05$).

2.2 Ethical considerations

Ethical approval was obtained from the Human Research and Ethics Committee (HREC) of Lagos University Teaching Hospital Idi-Araba (approval number: ADM/DCST/HREC/APP/5027). Permission to access donor registers was obtained from the respective institutions. As only secondary data was used, consent to participate was not required. The data was used solely for research purposes, and participant confidentiality was maintained throughout the study. The study was conducted ethically and adhered to the Declaration of Helsinki.

3 Results

A total of 47,473 donors were registered, of whom 11,484 (24.2%) were deferred. Of the 11,484 deferred donors, 6,601 were male, and 4,883 were female. The deferral rate was significantly higher among women (37.4% versus 19.2%) ($P < 0.001$). (Fig. 1)

The deferred donors were classified based on sex, age bracket (Table 1), and temporary or permanent deferral. A total of 2,032 (17.7%) were permanent deferrals, whereas 9,452 (82.3%) were temporary deferrals (Table 2).

Transfusion-transmissible infections (TTIs) were the most common cause of permanent deferral, present in 90.3% of cases (Table 3). The most common TTI recorded was seropositivity for HBV (61.9%), followed by HIV (17.7%). However, the prevalence of TTIs obtained in this study is not representative of the population prevalence, as routine pre-donation TTI screening was not performed at all study sites, and the results of post-donation screening performed at all study sites were not included in the analysed data.

The most common reason for temporary deferral was low haemoglobin levels, followed by high blood pressure observed in 43.8% and 16.5% of temporary deferral cases, respectively (Table 3).

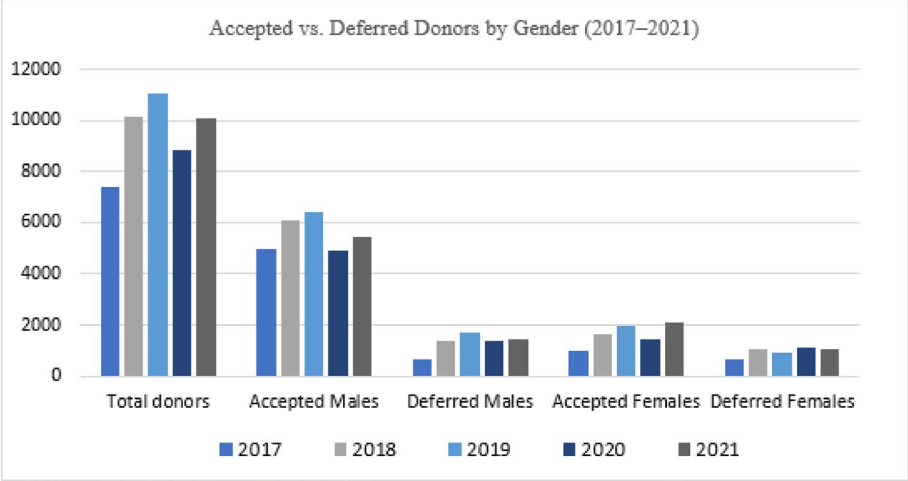


Fig. 1 Annual blood donor statistics: accepted vs. deferred donors by gender (2017–2021)

Table 1 Age and gender distributions of donors (accepted and deferred)

Variable	Registered donors	Number of donors accepted	Number of donors deferred	Acceptance rate (%)	Deferral rate (%)	Chi-square	*p* value
Sex							
Male	34,411 (72.5%)	27,810 (77.3%)	6601 (57.5%)	80.8	19.2	1710	< 0.0001
Female	13,062 (27.5%)	8179 (22.7%)	4883 (42.5%)	62.6	37.4		
Total	47,473 (100%)	35,989 (100%)	11,484 (100%)	75.8	24.2		
Age group (years)							
16–25	11,362 (23.9%)	8361 (23.2%)	3001 (26.1%)	73.6	26.4	1788.83	< 0.0001
26–35	14,149 (29.8%)	11,316 (31.4%)	2833 (24.7%)	80.0	20.0		
36–45	13,562 (28.6%)	10,688 (29.7%)	2874 (25.0%)	78.8	21.2		
46–55	6676 (14.1%)	4978 (13.8%)	1698 (14.8%)	74.6	25.4		
56–65	1527 (3.2%)	646 (1.8%)	881 (7.7%)	42.3	57.7		
> 65	197 (0.4%)	0 (0.00%)	197 (1.7%)	0.0	100.0		
Total	47,473 (100%)	35,989 (100%)	11,484 (100%)	75.8	24.2		

Table 2 Classification of blood donors according to type of deferral

Type of deferral	Number of donors	Percentage of total registered donors (%) *n* = 47,473	Chi-square	*p* value
Temporary	9452 (82.3%)	19.9	828.43	*< 0.0001*
Permanent	2032 (17.7%)	4.3		
Total deferral	11,484 (100%)	24.2		

Italics identifies *p* values < 0.05

Significant sex differences existed in most causes of donor deferral except overage, low BP, those on medication, peptic ulcer, jaundice, asthma, and recent vaccination (Table 4).

Low haemoglobin, high blood pressure, and underweight status were major factors across multiple age groups. Smoking and alcohol intake impacted younger age groups, particularly males (Tables 5 and 6).

Table 3 Causes of donor deferral (permanent and temporary)

Causes of deferral	Number of males deferred	Number of females deferred	Total deferred
Permanent causes*			
HBV	1213 (65.1%)	45 (26.8%)	1258 (61.9%)
HIV	344 (18.5%)	15 (8.9%)	359 (17.7%)
Overage	100 (5.4%)	97 (57.7%)	197 (9.7%)
HCV	136 (7.3%)	7 (4.2%)	143 (7.0%)
Syphilis	71 (3.8%)	4 (2.4%)	75 (3.7%)
Total	1864 (100.0%)	168 (100.00%)	2032 (100.0%)
Temporary causes**			
Low haemoglobin(< 12.5 g/dl)	1406 (37.3%)	2729 (48.0%)	4135 (43.8%)
High blood pressure	1044 (27.7%)	512 (9.0%)	1556 (16.5%)
Underweight (< 50 kg)	58 (1.5%)	567 (10.0%)	625 (6.6%)
Underage (< 18yrs)	207 (5.5%)	367 (6.5%)	574 (6.1%)
Low blood pressure	186 (4.9%)	231 (4.1%)	417 (4.4%)
Menstruation	0 (0.00%)	308 (5.4%)	308 (3.3%)
Medication	123 (3.3%)	154 (2.7%)	277 (2.9%)
Recent surgery	102 (2.7%)	144 (2.5%)	246 (2.6%)
Recent donation (< 3 months)	141 (3.7%)	63 (1.1%)	204 (2.2%)
Febrile illness	113 (3.0%)	86 (1.5%)	199 (2.1%)
Lactating mothers	0 (0.0%)	153 (2.7%)	153 (1.6%)
Recent tattoos (< 1 year)	94 (2.5%)	54 (1.0%)	148 (1.6%)
Recent piercing (< 1 year)	31 (0.8%)	90 (1.6%)	121 (1.3%)
Smoking in the past 24 h	106 (2.8%)	0 (0.0%)	106 (1.1%)
Recent alcohol intake	102 (2.7%)	0 (0.0%)	102 (1.1%)
Poor venous access	13 (0.3%)	83 (1.5%)	96 (1.0%)
Recent transfusion	22 (0.6%)	71 (1.3%)	93 (1.0%)
Pregnancy	0 (0.0%)	64 (1.1%)	64 (0.7%)
Peptic ulcer	11 (0.3%)	3 (0.1%)	1 (0.2%)
Jaundice	8 (0.2%)	2 (0.0%)	10 (0.1%)
Asthma	3 (0.1%)	0 (0.0%)	3 (0.0%)
Vaccination	1 (0.0%)	0 (0.0%)	1 (0.0%)
Total	3771 (100.0%)	5681 (100.0%)	9452 (100.0%)

*Statistical analysis (male vs. female) chi-square $\chi 2 = 1415.559.25$, $p < 0.0001$ (significant)

**Statistical analysis (male vs. female): chi-square: $\chi 2 = 385.961$, $p < 0.0001$ (significant)

4 Discussion

This study evaluated the causes and rates of blood donor rejection and deferral in Lagos, Southwest Nigeria. The prevalence of blood donor deferral in this study was 24.2%, and analysis of the causes of deferral revealed that most deferrals (82.3%) were due to temporary reasons ($p < 0.0001$). This is similar to other studies that reported temporary causes accounting for between 84% and 90.3% of deferred cases [12–16] but in contrast to the report by Valerian et al., who reported permanent causes as the most common cause of donor deferrals, accounting for 49.3% of deferrals [11].

Previous studies have reported different rates of donor deferrals. While several studies conducted in India reported relatively low deferral rates ranging from 2.6% to 6% [13, 14, 17], higher rates have been reported in other regions including 8.7% in Calabar, Southsouth Nigeria [18], 12.7% in northern Tanzania, and 16% in India and Uyo, Southsouth Nigeria [12, 19]. Deferral rates similar to those in this study were reported in Maiduguri, Northeast Nigeria, with a 17.7% temporary deferral rate [20], comparable with the 19.9% temporary deferrals in this study. The deferral rate observed in this study is slightly lower than the 32.5% reported in Nnewi, Southeast Nigeria [8].

Table 4 Causes and rates of donor derral

Causes of deferral	Male deferral %	Female deferral %	Total deferral %	Chi-square	*p*-value
Permanent causes					
HBV	3.5	0.3	2.6	1265.54	*< 0.0001*
HIV	1.0	0.1	0.8	322.21	*< 0.0001*
Overage	0.3	0.7	0.4	0.166	0.684 (not significant)
HCV	0.4	0.1	0.3	120.94	*< 0.0001*
Syphilis	0.2	0.0	0.2	60.99	*< 0.0001*
Total	5.4	1.3	4.3		
Temporary causes					
Low haemoglobin(< 12.5 g/dl)	4.1	20.9	8.7	585.92	*< 0.0001*
High blood pressure	3.0	3.9	3.3	233.21	*< 0.0001*
Underweight (< 50 kg)	0.2	4.3	1.3	417.02	*< 0.0001*
Underage (< 18yrs)	0.6	2.8	1.2	40.35	*< 0.0001*
Low blood pressure	0.5	1.8	0.9	3.27	0.071 (not significant)
Menstruation	0.0	2.4	0.6	302.89	*< 0.0001*
Medication	0.4	1.2	0.6	2.28	0.131 (not significant)
Recent surgery	0.3	1.1	0.5	5.51	*0.019*
Recent donation (< 3 months)	0.4	0.5	0.4	32.59	*< 0.0001*
Febrile illness	0.3	0.7	0.4	4.51	*0.034*
Lactating mothers	0.0	1.2	0.3	147.41	*< 0.0001*
Recent tattoos (< 1 year)	0.3	0.4	0.3	11.94	*0.0005*
Recent piercing (< 1 year)	0.1	0.7	0.3	25.96	*< 0.0001*
Smoking in the past 24 h	0.3	0.0	0.2	109	*< 0.0001*
Recent alcohol intake	0.3	0.0	0.2	104.78	*< 0.0001*
Poor venous access	0.0	0.6	0.2	47.47	*< 0.0001*
Recent transfusion	0.1	0.5	0.2	23.21	*< 0.0001*
Pregnancy	0.0	0.5	0.1	60.05	*< 0.0001*
Peptic ulcer	0.0	0.0	0.0	3.77	0.052 (not significant)
Jaundice	0.0	0.0	0.0	2.69	0.101 (not significant)
Asthma	0.0	0.0	0.0	1.41	0.235 (not significant)
Vaccination	0.0	0.0	0.0	0.0003	0.985 (not significant)
Total	11.0	43.5	19.9		

Italics identifies *p* values < 0.05

The differences in deferral rates observed between regions and even within regions can be explained by several factors. These include variations in donor eligibility or selection criteria, study protocols that either include or exclude TTIs, cultural attitudes towards blood donation, and educational and socioeconomic factors. A higher prevalence of donor deferral tends to occur in low-income countries than high-income countries due to the higher prevalence of anaemia in low-income countries [21].

The majority of the study subjects were males (72.5% versus 27.5%), with significantly higher deferral rates in female donors than in their male counterparts ($P < 0.0001$). This is consistent with previous studies in both low- and high-income countries, which reported higher deferral rates among the female population [8, 18, 19, 22, 23].

The number of female donors deferred for temporary reasons was significantly greater than that of male donors (60.1% versus 39.9%), which is attributable to the presence of sex-specific factors, such as menstruation, lactation, and pregnancy, among female subjects ($p < 0.0001$). The opposite was observed in permanent deferral, as male subjects dominated permanent deferred cases (91.7% versus 8.3%, $p < 0.0001$).

Table 5 Age and sex distribution and causes of temporary donor deferral

Causes/age	16–25 years		26–35 years		36–45 years		46–55 years		56–65 years		x2; *p*-value
Sex	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	
Low haemoglobin (< 12.5 g/dl)	475	781	307	634	294	626	199	435	131	253	12.595; *0.013*
High blood pressure	16	13	184	95	301	104	325	174	218	126	14.644; *0.006*
Underweight (< 50 kg)	46	271	11	201	1	45	0	37	0	13	22.392; *0.000*
Underage (< 18yrs)	207	367	0	0	0	0	0	0	0	0	-
Low blood pressure	49	71	51	91	51	32	27	23	8	14	16.948; *0.005*
Menstruation	0	53	0	123	0	132	0	0	0	0	-
Medication	15	12	21	34	27	53	31	40	29	15	14.159; *0.007*
Recent surgery	11	8	22	29	49	46	15	53	5	8	16.775; *0.002*
Recent donation (< 3 months)	32	9	54	33	30	19	20	1	5	1	12.268; *0.015*
Febrile illness	18	16	32	20	43	26	14	16	6	8	3.903; 0.419
Lactating mothers	0	5	0	72	0	76	0	0	0	0	-
Recent tattoos (< 1 year)	9	8	30	17	41	20	12	9	2	0	2.699; 0.609
Recent piercing (< 1 year)	4	13	11	28	14	40	2	9	0	0	-
Smoking in the past 24 h	3	0	23	0	51	0	23	0	6	0	-
Recent alcohol intake	4	0	31	0	44	0	17	0	6	0	-
Poor venous access	3	18	4	16	6	28	0	19	0	2	4.501; 0.342
Recent transfusion	0	0	5	13	9	28	6	18	2	12	-
Pregnancy	0	2	0	34	0	28	0	0	0	0	-
Peptic ulcer	2	0	4	0	4	2	1	1	0	0	-
Jaundice	3	2	3	0	2	0	0	0	0	0	-
Asthma	2	0	1	0	0	0	0	0	0	0	-
Vaccination	0	0	1	0	0	0	0	0	0	0	-
Total	899	1649	795	1440	967	1305	692	835	418	452	
% Total *n* = 9452	9.5	17.4	8.4	15.2	10.2	13.8	7.3	8.8	4.4	4.8	

Italics identifies *p* values < 0.05

In this study, the low haemoglobin level observed in 36.0% of the donors was the most common cause of donor rejection in both genders. These findings are similar to those of other studies [12, 13, 19, 24–26].

Anaemia was more common in females across all age groups, accounting for 66.0% of the cases deferred due to this cause. It was particularly high in females aged 16–35 years. This is likely linked to menstruation and nutritional factors. This reflects the fact that the incidence of clinical and subclinical anaemia is more pronounced in developing

Table 6 Age and sex distribution and causes of permanent donor deferral

Causes/age	16–25 years		26–35 years		36–45 years		46–55 years		56–65 years		>65 years	
SEX	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female
HBV	335	6	382	19	374	16	115	4	7	0	0	0
HIV	69	1	107	5	127	7	38	2	3	0	0	0
Over age	0	0	0	0	0	0	0	0	0	0	100	97
HCV	26	0	55	2	45	5	9	0	1	0	0	0
VDRL	16	0	27	1	25	3	3	0	0	0	0	0
Total	446	7	571	27	571	31	165	6	11	0	100	97
%Total $n = 2032$	21.9	0.3	28.1	1.3	28.1	1.5	8.1	0.3	0.5	0.0	4.9	4.8

Statistical analysis (male vs. female) chi-square $\chi2 = 1200.2$, $p < 0.0001$ (significant)

There is a significant association between causes and age groups in males ($\chi2 = 1100.5$, $p < 0.0001$) and in females ($\chi2 = 120.3$, $p < 0.0001$). The greater $\chi2$ in males suggests that the association between causes and age groups is stronger in males than in females

countries such as Nigeria due to inadequate nutrition and malaria endemicity, among others, and highlights the need for increased educational and nutritional awareness and nutritional supplementation, especially in females of reproductive age [19, 27].

High blood pressure (13.6%) and HBV seropositivity (11.0%) were the next leading causes of deferral. Other major causes of deferral recorded in this study are underweight (5.4%) and underage (5.0%).

High blood pressure was most prevalent in males aged 36–65 years, indicating age-related hypertension. Based on the records reviewed, the donor sites were unable to distinguish between first-time high blood pressure and known hypertensive subjects, as most of the donors had little or no regular medical checkups. Thus, high blood pressure was included under a temporary cause of deferral. Aneke et al. [8] and Chauhan et al. [28] also categorised high blood pressure as a cause of temporary deferral. However, other studies have categorised high blood pressure as a permanent cause of deferral [13, 14, 29, 30].

The most common cause of permanent donor deferrals was infection with HBV, followed by HIV, which was present in 11.0% and 3.1% of deferred donors, respectively. This pattern has also been reported in several studies [11, 23, 25, 28]. Unfortunately, the donor registers lacked data on the source of donors (family, replacement, paid, or voluntary). As such, authors could not make additional inferences on the effect of donor origin on deferral causes such as HBV positivity.

Concerning the age of the donors, deferral was highest in those aged 16–25 years (26.1%), followed by those aged 36–45 years (25.0%) and 26–35 years (24.7%). This is similar to findings by Kasraian and Negarestani, who reported the highest deferral rates among the 17–30 age groups [31]. The higher rejection rate observed in donors aged 16–25 years is due to high rates of anaemia (49.3%) and low body weight (12.4%). In addition, an appreciable percentage of donors within this age bracket were underage (22.5%). The highest prevalence of temporary deferral was observed in the 16–25 years age group, followed by the 36–45 and 26–35 years age groups (27.0%, 24,0%, and 23.7%, respectively), whereas the highest permanent deferral rate was observed in the 36–45 years age group (29.6%), followed by the 26–35 years age group (29.4%). This could indicate that most donors within that age bracket were sexually active and thus were more likely to contract TTIs. Similar findings were reported by Jethani N. et al., with a majority of permanently deferred cases seen among donors aged 31–40 years (32.3%) [14].

From this finding, it can be deduced that young people dominate temporary deferred cases. This young population represents a critical segment of the donor pool due to their generally better health and longer potential donation lifespan. Identifying and addressing the specific factors leading to their deferral, such as lifestyle choices, temporary health conditions, or lack of awareness, can significantly expand the pool of eligible donors. Tailored strategies, which include health screenings, education campaigns, and follow-up programs beginning at the donor clinics, can help young donors overcome temporary barriers, ensuring they remain engaged and eligible to contribute. Reducing deferral rates among this demographic increases the immediate donor pool and fosters a culture of sustained donation, securing a reliable supply of donors for years to come.

The majority of the causes of donor deferral documented in this study are short-term, preventable, and/or curable. This suggests that the level of societal knowledge and understanding regarding donor eligibility and selection criteria is poor. Increasing

awareness about donor eligibility criteria, risk factors for unsafe transfusion practices, and the importance of regular health checkups could help reduce deferral rates. It is essential to inform and explain to all deferred donors in detail the reasons for their deferral and the deferral period, and to counsel them to overcome the condition before their next visit. The LSBTS has deferral cards indicating reasons for deferral and the deferral period. This has proven effective in encouraging deferred donors to return after their deferral period, as it provides clear reasons and return timelines and helps deferred donors stay engaged. Expanding this system to include personalised counselling and follow-up reminders could improve donor recruitment and retention (see Fig. 1).

The high deferral rate observed in this study underscores the need for targeted public health interventions to address the leading causes of deferral, such as anaemia, HBV infection, and hypertension. Nutritional programs, particularly for females of reproductive age, could help reduce the prevalence of anaemia. Additionally, education campaigns aimed at reducing HBV and HIV transmission should be prioritized, especially among young adults who exhibit the highest deferral rates.

5 Conclusion

A high deferral rate reduces effective donations and contributes to blood shortages. Understanding the magnitude and pattern of donor deferrals provides the knowledge required to formulate policies, protocols, and processes to select donors, increase successful donations, reduce waste, enhance donor safety, and improve patient outcomes. There is a need to promote donor campaigns among relevant stakeholders. Without returning donors, there is no donor retention; hence, social interventions are needed to reduce nutritional anaemia and infectious disease transmissions.

6 Limitations

As a retrospective analysis, findings depended on the accuracy and completeness of recorded data. The dataset lacked stratification of first-time versus repeat donors, which precluded analysis of how deferral patterns may have differed between these groups. Additionally, the study population exhibited disproportionately low female representation (27.5%), which limited insights into sex-specific deferral causes. The absence of follow-up data further prevented assessment of whether deferred donors returned after their deferral period. Another constraint is the lack of donor source data (voluntary or family replacement), which could have revealed socioeconomic influences on deferral rates. Finally, the study did not evaluate the impact of donor education interventions, which could inform strategies to improve donor retention and blood supply sustainability.

Author contributions

O.A.O. Was involved in the study conceptualization, data curation, analysis, manuscript writing, review & editing. S.C.E. Drafted the methodology and was involved in data curation, analysis, validation, investigation, and writing the original draft of the manuscript. O.F.O. Supervised the project, and was involved in data analysis, manuscript review & editing. A.S.A. Supervised the project, and was involved in data analysis, manuscript review & editing. N.O.D. Was involved in drafting the methodology, data analysis, and manuscript review & editing. B.I.O. Was involved in the study conceptualization, investigation, and manuscript review & editing.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
Ethical approval with reference number ADM/DCST/HREC/APP/5027 was obtained from the Human Research and Ethics Committee (HREC) of the Lagos University Teaching Hospital Idi-Araba. Management approvals to access the donor registers were also obtained from the Lagos State Blood Transfusion Service (LSBTS), National Orthopaedic Hospital Igbobi, and General Hospital Mushin. As only secondary data was used, consent to participate was not required. The study adhered to the Declaration of Helsinki.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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